

Name : Sanika Jumde

Sem : V

PRN : 23070521225

(Regular Expressions)

Code:

```
RegularExpressionsExamples.html > ...
1  <!DOCTYPE html>
2  <head>
3  |   <title>Regular Expression Examples</title>
4  </head>
5  <body>
6  |   <script>
7  |       //Test if a string contains digits
8  |       let regex = /\d/;
9  |       console.log(regex.test("Hello 123"));
10 |       console.log(regex.test("Hello"));
11 |
12 |       //Validate Email Address
13 |       let emailRegex = /^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$/;
14 |       console.log(emailRegex.test("student@example.com")); // true
15 |       console.log(emailRegex.test("student@.com")); // false
16 |
17 |       //Find all words in a string
18 |       let text = "JavaScript is fun!";
19 |       let words = text.match(/\w+/g);
20 |       console.log(words); // ["JavaScript", "is", "fun"]
21 |
22 |       //Replace multiple spaces with a single space
23 |       let messy = "Hello World from JavaScript";
24 |       let clean = messy.replace(/\s+/g, " ");
25 |       console.log(clean); // "Hello World from JavaScript"
26 |   </script>
27 </body>
28 </html>
```

Output:

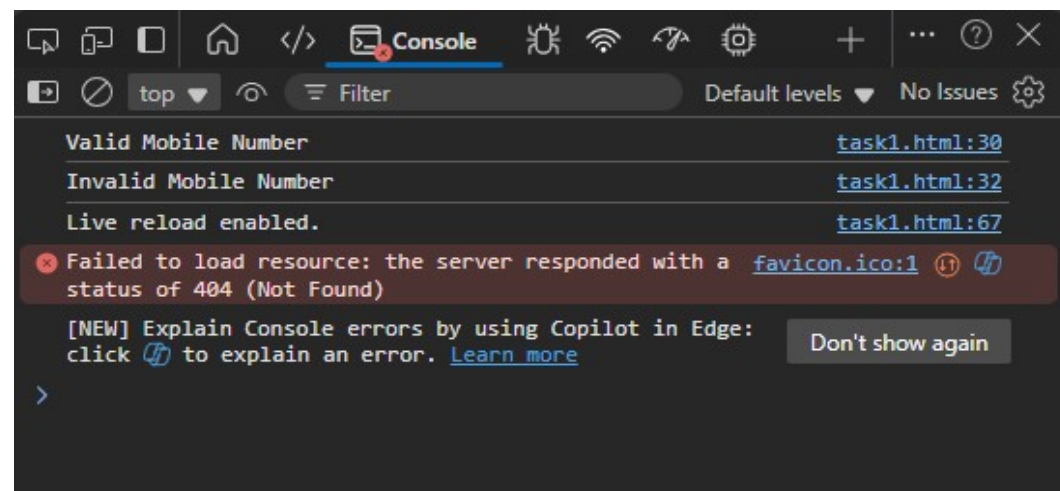
```
true                                     RegularExpressionsExamples.html:9
false                                    RegularExpressionsExamples.html:10
true                                     RegularExpressionsExamples.html:14
false                                    RegularExpressionsExamples.html:15
▼ Array(3) 1                             RegularExpressionsExamples.html:20
  0: "JavaScript"
  1: "is"
  2: "fun"
  length: 3
  ► [[Prototype]]: Array(0)
Hello World from JavaScript              RegularExpressionsExamples.html:25
Live reload enabled.                     RegularExpressionsExamples.html:55
Failed to load resource: the server responded with a status of 404 (Not Found)
[NEW] Explain Console errors by using Copilot in Edge:
click to explain an error. Learn more
Don't show again
```

Tasks:

Code 1:

```
task1.html > ...
1  <!-- Task 1: Validate Mobile Number
2  Create a regex to check if a number:
3
4  Starts with 6, 7, 8, or 9
5
6  Contains exactly 10 digits
7
8  Example Valid: 9876543210
9
10 Example Invalid: 1234567890
11
12 Hint:
13
14 let mobileRegex = /^[6-9]\d{9}$/;
15 Provide basic Javascript code for it.      -->
16
17 <!DOCTYPE html>
18 <head>
19   <title>Validate Mobile Number</title>
20 </head>
21 <body>
22   <script>
23     //Regex to validate mobile number
24     let mobileRegex = /^[6-9]\d{9}$/;
25
26     //Function to validate the number
27     function validateMobile(number)
28     {
29       if(mobileRegex.test(number))
30         console.log("Valid Mobile Number");
31       else
32         console.log("Invalid Mobile Number");
33     }
34
35     // Test cases
36     validateMobile("9876543210"); //Valid
37     validateMobile("1234567890"); //Invalid
38   </script>
39 </body>
40 </html>
```

Output:



Code 2:

```
<> task2.html > ...
1  <!-- Task 2: Extract Dates from Text
2  • Input: "Today's date is 12/09/2025 and tomorrow will be 13/09/2025."
3
4  • Output: ["12/09/2025", "13/09/2025"]
5
6  Hint:
7
8  let dateRegex = /\d{2}\d{2}\d{4}/g;
9  Provide basic Javascript code for it.      -->
10
11 <!DOCTYPE html>
12 <head>
13   <title>Extract Dates</title>
14 </head>
15 <body>
16   <script>
17     //Regex to extract dates in dd/mm/yyyy format from a passed string
18     let dateRegex = /\d{2}\d{2}\d{4}/g;
19
20     //Input text
21     let text = "Today's date is 12/09/2025 and tomorrow will be 13/09/2025.";
22
23     //Extract all matching dates
24     let dates = text.match(dateRegex);
25
26     //Display all extracted dates
27     console.log(dates);    //[ "12/09/2025", "13/09/2025" ]
28   </script>
29 </body>
30 </html>
```

Output:

task2.html:27

```
Array(2) 1
  0: "12/09/2025"
  1: "13/09/2025"
  length: 2
  [[Prototype]]: Array(0)
```

Live reload enabled. task2.html:57

Failed to load resource: the server responded with a status of 404 (Not Found) favicon.ico:1

[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. [Learn more](#) Don't show again

Code 3:

```
task3.html > ...
1  <!-- Task 3: Password Strength Checker
2  Rules:
3
4  • Minimum 8 characters
5
6  • At least 1 uppercase, 1 lowercase, 1 number, and 1 special character
7
8  Regex:
9
10 let passwordRegex = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$/;
11
12 Examples:
13
14 • "StrongPass1!" → Valid
15
16 • "weakpass" → Invalid
17 Provide basic javascript code for it -->
18
19 <!DOCTYPE html>
20 <head>
21 |   <title>Password Strength Checker</title>
22 </head>
23 <body>
24 |   <script>
25 |     //Regex to validate strong password
26 |     let passwordRegex = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@$!%*?&])[A-Za-z\d@$!%*?&]{8,}$/;
27
28 |     //Function to check password strength
29 |     function checkPassword(password)
30 |     {
31 |       if(passwordRegex.test(password))
32 |       {
33 |         console.log("Strong Password");
34 |       }
35 |       else
36 |       {
37 |         console.log("Weak Password");
38 |       }
39 |     }
40 |
41 |     //Test cases
42 |     checkPassword("StrongPass1!"); //Strong Password
43 |     checkPassword("weakpass");    //Weak Password
44 |
45 |   </script>
46 </body>
47 </html>
```

Output:

Strong Password [task3.html:32](#)

Weak Password [task3.html:34](#)

Live reload enabled. [task3.html:69](#)

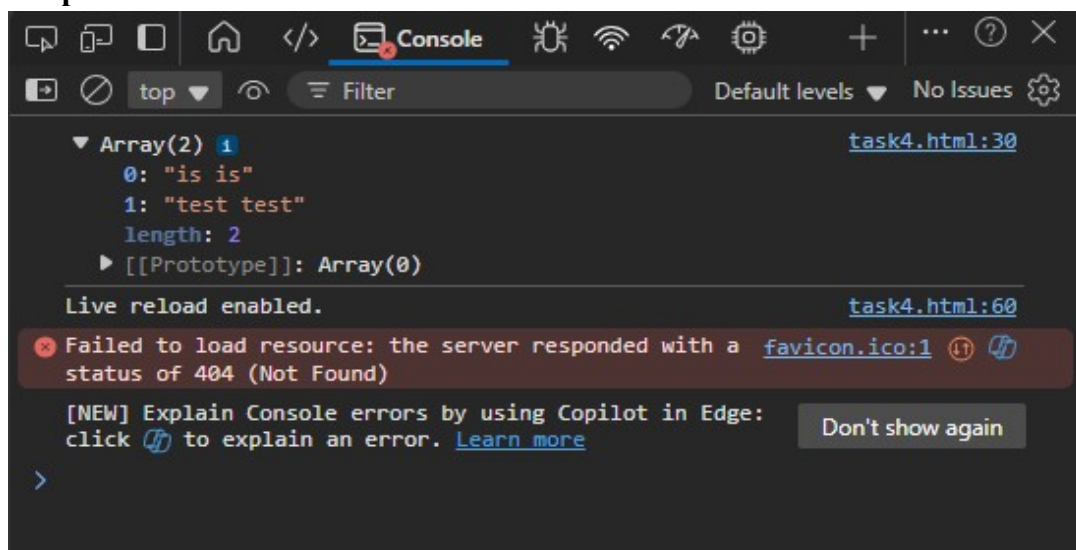
Failed to load resource: the server responded with a status of 404 (Not Found) [favicon.ico:1](#)

[NEW] Explain Console errors by using Copilot in Edge: click to explain an error. [Learn more](#) [Don't show again](#)

Code 4:

```
<> task4.html > ...
1  <!-- Task 4: Find Duplicate Words
2  • Input: "This is is a test test"
3
4  • Output: ["is", "test"]
5
6  Hint:
7
8  let text = "This is is a test test";
9
10 let dupRegex = /\b(\w+)\s+\1\b/g;
11
12 console.log(text.match(dupRegex)); // ["is", "test"]      -->
13
14 <!DOCTYPE html>
15 <head>
16   <title>Display Duplicate Words</title>
17 </head>
18 <body>
19   <script>
20     //Input text
21     let text = "This is is a test test";
22
23     //Regex to find duplicate consecutive words
24     let dupRegex = /\b(\w+)\s+\1\b/g;
25
26     //Match duplicates
27     let duplicates = text.match(dupRegex);
28
29     //Displaying duplicate values
30     console.log(duplicates); //["is", "test"]
31   </script>
32 </body>
33 </html>
```

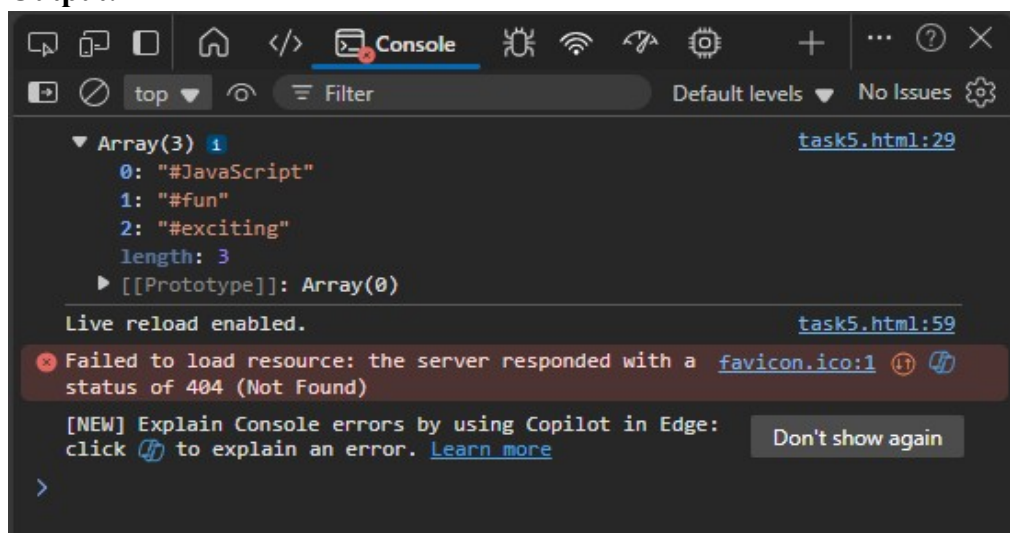
Output:



Code 5:

```
<> task5.html > ...
1  <!-- Task 5: Extract Hashtags from a Tweet
2  • Input: "Learning #JavaScript is #fun and #exciting!"
3  • Output: ["#JavaScript", "#fun", "#exciting"]
4  Hint:
5
6  let tweet = "Learning #JavaScript is #fun and #exciting!";
7
8  let hashtags = tweet.match(/#[A-Za-z0-9_]+/g);
9
10 console.log(hashtags);
11 Provide basic javascript code for it is      -->
12
13 <!DOCTYPE html>
14 <head>
15 |   <title>Extract Hashtags</title>
16 </head>
17 <body>
18 |   <script>
19 |     //Input tweet
20 |     let tweet = "Learning #JavaScript is #fun and #exciting!";
21 |
22 |     //Regex to find hashtags
23 |     let hashtagRegex = /#[A-Za-z0-9_]+/g;
24 |
25 |     //Extract all hashtags
26 |     let hashtags = tweet.match(hashtagRegex);
27 |
28 |     //Display all extracted #tag words
29 |     console.log(hashtags);      //["#JavaScript", "#fun", "#exciting"]
30 |   </script>
31 </body>
32 </html>
```

Output:



The screenshot shows a web browser's developer console with the following content:

- Top bar: Includes icons for various developer tools and a search bar.
- Filter: Set to "top".
- Default levels: Set to "Default levels".
- No Issues: A button to toggle the display of issues.
- Array(3) 1: The output of the console.log statement, showing an array of three elements: "#JavaScript", "#fun", and "#exciting". The array has a length of 3.
- Live reload enabled: A message indicating that live reload is enabled.
- Failed to load resource: the server responded with a status of 404 (Not Found): An error message indicating that a resource (favicon.ico) was not found.
- [NEW] Explain Console errors by using Copilot in Edge: A message suggesting the use of Copilot to explain errors.
- Don't show again: A button to dismiss the message.

Code 6:

Problem Set:

Code:

```
<> practiceProblems.html > ...
1  <!DOCTYPE html>
2  <head>
3    <title>Practice Problems</title>
4  </head>
5  <body>
6    <script>
7      //Validate PIN(6 digits)
8      let pin = "440001";
9      let pinRegex = /^d{6}$/;
10
11     if(pinRegex.test(pin))
12       console.log("Valid");
13     else
14       console.log("Invalid");
15
16     //Validate URL
17     let url = "https://google.com";
18     let urlRegex = /^(https?:\/\/\w[-]+\.[a-z]{2,})$/i;
19
20     if(urlRegex.test(url))
21       console.log("Valid");
22     else
23       console.log("Invalid");
24
25     //Extract Numbers
26     let text1 = "Order 123 costs $45";
27     let numberRegex = /\d+/g;
28
29     let numbers = text1.match(numberRegex);
30     console.log(numbers); //["123", "45"]
31
32     //Extract Emails
33     let text2 = "Contact us at help@abc.com or sales@xyz.org";
34     let emailRegex = /\w[-]+\.[a-z]{2,}/g;
35
36     let emails = text2.match(emailRegex);
37     console.log(emails); //["help@abc.com", "sales@xyz.org"]
38   </script>
39 </body>
40 </html>
```

Output:

Code 7:

